

Fractional Quantile Transmutation, Complex-Order Resolvents, and Nonlocal Rvachev Calculus

New integral and transform frontiers for F , F^{-1} , and up

Research report prepared for Vladimir Reshetnikov

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Abstract

Let F be the Fabius distribution function on $[0, 1]$, let $G = F^{-1}$, and let up be Rvachev's compactly supported even density. This report develops an operator-level calculus for integrals and fractional derivatives involving all three functions. Its main result is a quantile–Volterra transmutation: for arbitrary complex integration order β with $\Re\beta > 0$, a Riemann–Liouville integral of $\phi \circ G$ is converted into an ordinary integral with kernel $(u - F(x))^\beta$. The formula extends entirely in a power parameter, thereby covering positive, negative, and complex powers of G , as well as logarithmic powers obtained by parameter differentiation. A parallel transmutation is proved for Caputo derivatives of every noninteger order; the compact differential operator $\mathcal{L}_F = (F')^{-1} \frac{d}{dx}$ packages all higher inverse-function Bell polynomials.

A two-parameter generalized Stieltjes transform $\mathcal{S}_\lambda(z) = \int_0^1 (z - x)^{-\lambda} dF(x)$ is then developed. It satisfies a dyadic differential equation, an order-lowering refinement identity, a moment expansion, an exact fractional order-shift calculus in z , and a boundary-jump formula that recovers a right-sided fractional integral of F' . For Rvachev's function, Weyl and Riesz fractional derivatives obey an exact order-lowering dilation equation. Unlike ordinary derivatives, noninteger derivatives destroy compact support; outside $[-1, 1]$ their algebraic tails have convergent expansions whose coefficients are the exact rational even moments of up .

Finally, if $D = X_2 - X_1$ for independent Fabius variables, the normalized positive-part transform $\mathcal{K}_\beta(a) = \mathbb{E}(D - a)_+^\beta / \Gamma(\beta + 1)$ extends to an entire function of the order β at every interior threshold. Positive orders are fractional stop-loss integrals, order 0 is a tail probability, order -1 is the autocorrelation density, and negative integers are its derivatives. The whole hierarchy satisfies a triangular dyadic refinement equation and a fractional Fourier-energy identity. Numerical experiments reconstructing up from its infinite sinc product verify the new identities to between roughly 10^{-9} and 10^{-13} . The report distinguishes proved results, conditional endpoint refinements, and conjectural research directions, and records its novelty only relative to the audited repository snapshot rather than claiming exhaustive global priority.

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1 Scope, corpus audit, and novelty boundary

1.1 Repository snapshot

The requested corpus was the recursive tree `Analysis/FabiusFunction/docs` in Vladimir Reshetnikov’s `ProveIt` repository [11]. The audit was pinned to commit `22f3939f162b8fe9481b4affcd6ea12e7f45e8fd`; pinning matters because the branch moved during the audit. GitHub’s indexed code search returned 84 `.tex` sources under this tree. They fall into four broad classes:

1. the principal human-readable exposition and mathematical glossary;
2. proved-result notes tied closely to the Lean development;
3. non-formalized frontier reports and successive drafts;
4. imported papers, translations, and OCR-corrected literature used as background sources.

All paths were recursively inventoried and searched for the terms relevant to integrals, transforms, inverse functions, Mellin and Fourier analysis, negative powers, and fractional calculus. Earlier versions and imported copies were de-duplicated by path, title, and mathematical content. The most important overlap checks were made against the two newest broad syntheses:

- `non-formalized-research-frontiers/drafts/Fabius_Integral_and_Transform_Frontiers/fabius_integrals_and_transforms_frontier_report.tex`;
- `non-formalized-research-frontiers/drafts/fabius_integral_frontiers_bundle/fabius_integral_frontiers.tex`.

These already subsume much of the material in the monomial-antiderivative, dyadic primitive-ladder, derivative-norm, inverse-beta, and Mellin-frontier reports. Consequently, this report does not spend pages reproving the standard polynomial primitive ladder or reproducing tables already present there.

1.2 What the repository already contained

The following topics were treated as part of the prior-art boundary for this snapshot:

- ordinary and iterated primitives of F , $1 - F$, and up , including monomial, polynomial, complex-power, and several negative-power weights;
- dyadic reduction of primitives and exact moment recurrences;
- causal fractional integrals of up , Mellin transforms, and moment-tail expansions;
- inverse-Fabius integer layer-cake formulas, beta/Jacobi transforms, and inverse moments;
- the ordinary Cauchy–Stieltjes transform and its basic dyadic equation;
- derivative norms, Fourier–Parseval formulas, and basic autocorrelation identities;
- endpoint expansions expressed through lower Lambert W , including log-periodic corrections;
- the warning that a naive single-dilation fractional analogue of the integer derivative equation is obstructed by the dyadic log-periodic phase.

The present work uses those facts as infrastructure. In particular, it does not claim novelty for the probabilistic representation, the sinc product, the moment recurrences, or the endpoint Lambert structure.

1.3 Results that are new relative to the audited snapshot

The novelty comparison found no existing occurrence of the following operator systems in the repository:

1. the arbitrary-order quantile–Volterra transmutation (25), including its entire continuation in a power parameter and the resulting logarithmic identities;
2. the all-orders Caputo transmutation through $\mathcal{L}_F = (F')^{-1}D_x$;
3. the generalized Stieltjes family $\mathcal{S}_\lambda(z)$ as a coupled complex-order/dyadic system, including fractional order shifts in the spectral variable and fractional boundary inversion;
4. the exact Weyl order-lowering dilation law and the rational-moment algebraic tails of Weyl and Riesz derivatives of up;
5. the entire order hierarchy $\mathcal{K}_\beta(a)$ for Fabius differences, with negative orders equal to autocorrelation derivatives and with a triangular refinement equation valid by analytic continuation.

Normalization warning 1.1. The word “new” below always means “not found in the audited repository at the pinned commit.” The external literature on fractional transforms is large, and no absolute priority claim is made. The formulas are nevertheless derived from first principles here and are suitable for independent checking or formalization.

2 Common probabilistic and analytic model

2.1 The three functions

We use the standard probabilistic realization of the Fabius distribution [5, 1]. Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[0, 1]$, and define

$$X := \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad Y := 2X - 1. \quad (1)$$

Then $X \in [0, 1]$, $Y \in [-1, 1]$, and

$$F(x) := \mathbb{P}\{X \leq x\}, \quad G(u) := F^{-1}(u), \quad \mathbb{P}\{Y \in dy\} = \text{up}(y) dy. \quad (2)$$

The normalizations used throughout are

$$F'(x) = 2 \text{up}(2x - 1), \quad 0 < x < 1, \quad (3)$$

$$F(1 - x) = 1 - F(x), \quad G(1 - u) = 1 - G(u), \quad (4)$$

$$\text{up}(-y) = \text{up}(y), \quad \int_{-1}^1 \text{up}(y) dy = 1. \quad (5)$$

In particular,

$$F(x) = \int_{-1}^{2x-1} \text{up}(y) dy, \quad \text{up}(y) = \begin{cases} F(1+y), & -1 \leq y \leq 0, \\ F(1-y), & 0 \leq y \leq 1, \\ 0, & |y| \geq 1. \end{cases} \quad (6)$$

The random series has the fixed-point forms

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad Y \stackrel{d}{=} \frac{V + Y'}{2}, \quad (7)$$

where $U \sim \text{Unif}[0, 1]$, $V \sim \text{Unif}[-1, 1]$, and the primed variables are independent copies. This random-series and infinite-product view goes back to Jessen–Wintner and its later Fabius/Rvachev formulations [9, 5, 1]. With Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-i\xi x} f(x) dx$,

$$\widehat{\text{up}}(\xi) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{\xi}{2^j}\right), \quad \text{sinc } z := \frac{\sin z}{z}. \quad (8)$$

2.2 Quantile and density dictionaries

For every integrable test function ψ ,

$$\int_0^1 \psi(G(u)) du = \int_0^1 \psi(x) dF(x) = \int_{-1}^1 \psi\left(\frac{1+y}{2}\right) \text{up}(y) dy. \quad (9)$$

This simple identity is the common source of the inverse-function, Stieltjes, and autocorrelation formulas below. It gives, for example,

$$d_n := \mathbb{E}X^n = \int_0^1 G(u)^n du, \quad c_n := \mathbb{E}Y^{2n} = \int_{-1}^1 y^{2n} \text{up}(y) dy. \quad (10)$$

The fixed-point equations yield exact rational recurrences; related moment recurrences are developed in [1, 8]:

$$d_0 = 1, \quad (2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{d_k}{n - k + 1}, \quad (11)$$

$$c_0 = 1, \quad c_n = \frac{1}{(2n+1)(4^n - 1)} \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \quad (12)$$

Thus

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}, \dots \quad (13)$$

The value $c_2 = 19/675$ is important in the nonlocal tail expansions; replacing it by $1/75$ would correspond to a different and incorrect normalization.

2.3 Fractional-calculus conventions

We follow the standard Riemann–Liouville, Caputo, Weyl, and Riesz normalizations of [4, 12, 10]. For $a < u$, $\Re\beta > 0$, and locally integrable h , the left Riemann–Liouville integral is

$$(I_{a+}^{\beta} h)(u) := \frac{1}{\Gamma(\beta)} \int_a^u (u-v)^{\beta-1} h(v) dv. \quad (14)$$

For $m-1 < \alpha < m$, $m \in \mathbb{N}$, the Caputo derivative is

$$({}^C D_{a+}^{\alpha} h)(u) := \frac{1}{\Gamma(m-\alpha)} \int_a^u (u-v)^{m-\alpha-1} h^{(m)}(v) dv. \quad (15)$$

For $0 < \alpha < 1$, this reduces to the familiar integral of h' . Right-sided operators use the standard sign $(-1)^m$.

On the whole line, let $k_{\sigma}(x) = x_{+}^{\sigma-1}/\Gamma(\sigma)$ denote the standard Gamma-normalized causal homogeneous distribution, continued entirely in σ . The left Weyl operator is $D_{+}^{\alpha} f := k_{-\alpha} * f$. Wherever the multiplier is an ordinary function, and elsewhere in the corresponding boundary-value distributional sense,

$$\widehat{D_{+}^{\alpha} f}(\xi) = (i\xi)^{\alpha} \widehat{f}(\xi), \quad (16)$$

with the branch obtained from the causal kernel. This normalization removes zero-frequency polynomial ambiguities and makes the order family entire. The Riesz fractional Laplacian is

$$(-\Delta)^{\alpha/2} f(\xi) = |\xi|^\alpha \widehat{f}(\xi). \quad (17)$$

Different fractional derivatives answer different boundary-value questions; no formula below silently identifies Caputo, Riemann–Liouville, Weyl, and Riesz operators.

3 An entire-order primitive ladder

Before turning to the genuinely new inverse-function formulas, it is useful to package ordinary derivatives and antiderivatives in a normalization that continues through every integer order.

3.1 Fabius distribution ladder

For $x \in (0, 1)$ and initially $\Re\beta > -1$, define

$$\mathfrak{F}_\beta(x) := \frac{1}{\Gamma(\beta+1)} \int_0^x (x-t)^\beta F'(t) dt. \quad (18)$$

For $\beta > 0$, this is $I_{0+}^\beta F(x)$; at $\beta = 0$ it is $F(x)$.

Proposition 3.1 (Entire order continuation). *For every fixed $x \in (0, 1)$, $\beta \mapsto \mathfrak{F}_\beta(x)$ extends to an entire function. It obeys*

$$\frac{\partial}{\partial x} \mathfrak{F}_\beta(x) = \mathfrak{F}_{\beta-1}(x) \quad (19)$$

for all $\beta \in \mathbb{C}$, and at negative integers

$$\mathfrak{F}_{-m}(x) = F^{(m)}(x), \quad m \geq 1. \quad (20)$$

Proof. Write $s = x - t$. Because F' and all of its derivatives vanish at 0, repeated integration by parts gives, for every $N \geq 0$ and $\Re\beta > -N - 1$,

$$\mathfrak{F}_\beta(x) = \frac{1}{\Gamma(\beta+N+1)} \int_0^x s^{\beta+N} F^{(N+1)}(x-s) ds. \quad (21)$$

The formulas agree on overlaps and cover the whole β -plane. The reciprocal Gamma factor cancels the Mellin poles at the nonpositive integers. Differentiation proves (19); substituting $\beta = -m$ in (21) with $N = m$ gives $F^{(m)}(x) - F^{(m)}(0) = F^{(m)}(x)$. $\square$

3.2 Rvachev ladder and exterior moment series

For $x > -1$ and initially $\Re\sigma > 0$, put

$$\mathfrak{U}_\sigma(x) := \frac{1}{\Gamma(\sigma)} \int_{-1}^{\min(x,1)} (x-t)^{\sigma-1} \text{up}(t) dt. \quad (22)$$

For $-1 < x < 1$, flatness at -1 again gives an entire continuation in σ , with

$$\partial_x \mathfrak{U}_\sigma = \mathfrak{U}_{\sigma-1}, \quad \mathfrak{U}_0 = \text{up}, \quad \mathfrak{U}_{-m} = \text{up}^{(m)}. \quad (23)$$

For $x > 1$, the binomial series converges absolutely and yields

$$\mathfrak{U}_\sigma(x) = \frac{x^{\sigma-1}}{\Gamma(\sigma)} \sum_{j=0}^{\infty} \frac{(1-\sigma)_{2j}}{(2j)!} \frac{c_j}{x^{2j}}. \quad (24)$$

The Weyl tail formula of Section 7 is precisely the specialization $\sigma = -\alpha$ of this exterior ladder. Thus ordinary antiderivatives, ordinary derivatives, and nonlocal fractional tails are not separate phenomena; they are different slices of one Gamma-normalized order family.

4 Quantile–Volterra transmutation

4.1 The master theorem

The basic obstacle in integrating $G = F^{-1}$ is the derivative $G'(u) = 1/F'(G(u))$, which becomes very large at both endpoints. A layer-cake rearrangement removes this derivative completely.

Theorem 4.1 (Quantile–Volterra transmutation). *Let $0 \leq a < u < 1$, put $q_a = G(a)$ and $q = G(u)$, let $\Re\beta > 0$, and let ϕ be absolutely continuous on $[q_a, q]$. Then*

$$I_{a+}^{\beta}[\phi(G) - \phi(q_a)](u) = \frac{1}{\Gamma(\beta + 1)} \int_{q_a}^q \phi'(x)(u - F(x))^{\beta} dx. \quad (25)$$

For $a = 0$ and $\phi(0) = 0$,

$$I_{0+}^{\beta}(\phi \circ G)(u) = \frac{1}{\Gamma(\beta + 1)} \int_0^{G(u)} \phi'(x)(u - F(x))^{\beta} dx. \quad (26)$$

Proof. For $v \in [a, u]$,

$$\phi(G(v)) - \phi(q_a) = \int_{q_a}^{G(v)} \phi'(x) dx \quad (27)$$

$$= \int_{q_a}^q \phi'(x) \mathbf{1}_{\{F(x) \leq v\}} dx. \quad (28)$$

Fubini's theorem is immediate for nonnegative ϕ' and follows for an arbitrary absolutely continuous ϕ by applying the same argument to its positive and negative variations. Hence

$$I_{a+}^{\beta}[\phi(G) - \phi(q_a)](u) = \frac{1}{\Gamma(\beta)} \int_{q_a}^q \phi'(x) \int_{F(x)}^u (u - v)^{\beta-1} dv dx \quad (29)$$

$$= \frac{1}{\Gamma(\beta + 1)} \int_{q_a}^q \phi'(x)(u - F(x))^{\beta} dx. \quad (30)$$

For complex β , the kernel is positive real before exponentiation, so there is no branch ambiguity. $\square$

The theorem trades an integral over probability level v containing the inverse $G(v)$ for an integral over physical position x containing only the forward Fabius function. The singular inverse derivative never appears. For $\beta = n \in \mathbb{N}$, it gives every n -fold antiderivative at once; for noninteger β , the same formula is a genuine fractional primitive.

4.2 Powers, negative powers, and an entire parameter

Taking $\phi(x) = x^p$ gives, initially for $\Re p > 0$,

$$I_{0+}^{\beta} G^p(u) = \frac{p}{\Gamma(\beta + 1)} \int_0^q x^{p-1} (u - F(x))^{\beta} dx, \quad q = G(u). \quad (31)$$

The endpoint flatness of F permits much more.

Theorem 4.2 (Entire continuation in the power). *Fix $u \in (0, 1)$ and $\Re\beta > 0$. The function $p \mapsto I_{0+}^{\beta} G^p(u)$, initially represented by (31), extends to an entire function of $p \in \mathbb{C}$:*

$$I_{0+}^{\beta} G^p(u) = \frac{1}{\Gamma(\beta + 1)} \left[u^{\beta} q^p + p \int_0^q x^{p-1} \left((u - F(x))^{\beta} - u^{\beta} \right) dx \right]. \quad (32)$$

Here $q^p = \exp(p \log q)$ with the real logarithm.

Proof. Add and subtract u^β inside (31); for $\Re p > 0$, $p \int_0^q x^{p-1} u^\beta dx = u^\beta q^p$. Near $x = 0$,

$$(u - F(x))^\beta - u^\beta = -\beta u^{\beta-1} F(x) + \mathcal{O}(F(x)^2). \quad (33)$$

Since $F(x)$ is flat at zero, the regularized Mellin integral converges locally uniformly for every $p \in \mathbb{C}$. It is therefore entire, as is q^p . $\square$

This formula gives convergent representations for every negative and complex power. It also explains why negative inverse moments are much better behaved here than for an algebraic endpoint distribution: the superpolynomial flatness of F defeats every pole x^{p-1} .

4.3 Logarithmic powers and exponential generating functions

Differentiate (32) at $p = 0$.

Corollary 4.3 (Fractional logarithmic moments). *For every integer $m \geq 1$,*

$$I_{0+}^\beta [(\log G)^m](u) = \frac{1}{\Gamma(\beta + 1)} \left[u^\beta (\log q)^m \right. \quad (34)$$

$$\left. + m \int_0^q \frac{(\log x)^{m-1}}{x} \left((u - F(x))^\beta - u^\beta \right) dx \right]. \quad (35)$$

Likewise, with $\phi(x) = e^{sx}$,

$$I_{0+}^\beta (e^{sG} - 1)(u) = \frac{s}{\Gamma(\beta + 1)} \int_0^q e^{sx} (u - F(x))^\beta dx. \quad (36)$$

Expanding either side in s simultaneously generates all positive integer powers. Differentiating with respect to s at nonzero values generates exponentially tilted inverse moments.

4.4 Definite integrals and fractional order statistics

At $u = 1$, (26) becomes

$$I_{0+}^\beta (\phi \circ G)(1) = \frac{1}{\Gamma(\beta + 1)} \int_0^1 \phi'(x) (1 - F(x))^\beta dx. \quad (37)$$

For $\phi(x) = x$ and symmetry,

$$I_{0+}^\beta G(1) = \frac{1}{\Gamma(\beta + 1)} \int_0^1 F(x)^\beta dx. \quad (38)$$

When $\beta = n \in \mathbb{N}$, if $M_n = \max(X_1, \dots, X_n)$, then $\mathbb{P}(M_n \leq x) = F(x)^n$ and

$$n! I_{0+}^n G(1) = \int_0^1 F(x)^n dx = \mathbb{E}(1 - M_n). \quad (39)$$

Thus the integration order continuously interpolates the expected gap above a maximum of an integer number of samples. Formula (38) provides a natural “fractional sample-size” extension without inventing a noninteger collection of random variables. For $n = 1$, it gives $\int_0^1 G(u) du = 1/2$.

More generally,

$$I_{0+}^\beta G^p(1) = \frac{p}{\Gamma(\beta + 1)} \int_0^1 x^{p-1} (1 - F(x))^\beta dx, \quad (40)$$

with entire continuation in p supplied by (32). This is a two-parameter bridge between inverse moments and beta-like powers of the forward distribution.

4.5 Log-convexity and Turan inequalities in the order

Suppose $\phi' \geq 0$. For real $\beta > -1$, define the direct-kernel order transform

$$T_\beta(u) := \int_{q_a}^q \phi'(x)(u - F(x))^\beta dx. \quad (41)$$

For $\beta > 0$ this equals $\Gamma(\beta + 1)I_{a+}^\beta[\phi(G) - \phi(q_a)](u)$ by [Theorem 4.1](#); the direct integral also makes sense on the larger interval $\beta > -1$.

Proposition 4.4 (Order log-convexity). *For real $\beta > -1$ in the interval of convergence, $T_\beta(u)$ is log-convex. In particular, whenever all three terms exist,*

$$T_\beta(u)^2 \leq T_{\beta-h}(u)T_{\beta+h}(u). \quad (42)$$

If $M = \phi(q) - \phi(q_a) > 0$, then

$$\left(\frac{T_\beta(u)}{M} \right)^{1/\beta} \text{ is nondecreasing for } \beta > 0. \quad (43)$$

Proof. Normalize $\phi'(x) dx$ to a probability measure and apply Holder's inequality to the positive random variable $u - F(x)$. The second assertion is the monotonicity of power means. $\square$

The Turan inequality is potentially useful for obtaining rigorous bounds on fractional primitives from adjacent integer orders already known exactly or recursively.

4.6 Semigroup identity after transmutation

The Riemann–Liouville semigroup $I^\gamma I^\beta = I^{\beta+\gamma}$ becomes the nontrivial forward-function identity

$$\begin{aligned} I_{a+}^\gamma \left[\frac{1}{\Gamma(\beta + 1)} \int_{q_a}^{G(v)} \phi'(x)(v - F(x))^\beta dx \right] (u) \\ = \frac{1}{\Gamma(\beta + \gamma + 1)} \int_{q_a}^q \phi'(x)(u - F(x))^{\beta+\gamma} dx. \end{aligned} \quad (44)$$

This is immediate from the theorem but useful computationally: two nested fractional integrations in the quantile variable collapse to one forward Fabius kernel.

5 Caputo derivatives of the inverse Fabius function

5.1 Orders between zero and one

Theorem 5.1 (Caputo quantile transmutation). *Let $0 < \alpha < 1$, $0 \leq a < u < 1$, and let ϕ be continuously differentiable on $[G(a), G(u)]$. Then*

$${}^C D_{a+}^\alpha (\phi \circ G)(u) = \frac{1}{\Gamma(1 - \alpha)} \int_{G(a)}^{G(u)} \phi'(x)(u - F(x))^{-\alpha} dx. \quad (45)$$

Whenever the endpoint integral converges, $a = 0$ is allowed.

Proof. By definition,

$${}^C D_{a+}^\alpha (\phi \circ G)(u) = \frac{1}{\Gamma(1 - \alpha)} \int_a^u (u - v)^{-\alpha} \phi'(G(v))G'(v) dv. \quad (46)$$

Set $v = F(x)$. Since $G'(F(x))F'(x) = 1$, the inverse derivative cancels exactly, producing (45). $\square$

For powers,

$${}^C D_{0+}^\alpha G^p(u) = \frac{p}{\Gamma(1-\alpha)} \int_0^q x^{p-1} (u - F(x))^{-\alpha} dx, \quad \Re p > 0. \quad (47)$$

For $p = 1$,

$${}^C D_{0+}^\alpha G(u) = \frac{1}{\Gamma(1-\alpha)} \int_0^{G(u)} (u - F(x))^{-\alpha} dx. \quad (48)$$

The singularity at $x = G(u)$ is only of order $(G(u) - x)^{-\alpha}$ and is therefore integrable.

With the standard right-Caputo sign convention and the symmetry of G ,

$${}^C D_{1-}^\alpha G(u) = -{}^C D_{0+}^\alpha G(1 - u). \quad (49)$$

5.2 Every noninteger Caputo order

The cancellation in [Theorem 5.1](#) is the first member of a larger hierarchy. Define the first-order differential operator

$$\mathcal{L}_F := \frac{1}{F'(x)} \frac{d}{dx}. \quad (50)$$

It is differentiation with respect to probability level, pulled back to the x coordinate.

Lemma 5.2 (Inverse Bell polynomials in operator form). *For every integer $m \geq 0$ and every smooth ϕ ,*

$$\left. \frac{d^m}{du^m} \phi(G(u)) \right|_{u=F(x)} = \mathcal{L}_F^m \phi(x). \quad (51)$$

Proof. The case $m = 1$ is the inverse-function rule. If the formula holds at m , then differentiation in u at $u = F(x)$ is exactly the operation $(F'(x))^{-1} D_x = \mathcal{L}_F$. $\square$

The first three cases are

$$\mathcal{L}_F \phi = \frac{\phi'}{F'}, \quad (52)$$

$$\mathcal{L}_F^2 \phi = \frac{\phi''}{(F')^2} - \frac{\phi' F''}{(F')^3}, \quad (53)$$

$$\mathcal{L}_F^3 \phi = \frac{\phi'''}{(F')^3} - \frac{3\phi'' F'' + \phi' F'''}{(F')^4} + \frac{3\phi' (F'')^2}{(F')^5}. \quad (54)$$

These are precisely the inverse-function Bell-polynomial expressions, but the operator notation scales to arbitrary order without combinatorial clutter.

Theorem 5.3 (All-orders Caputo transmutation). *Let $m - 1 < \alpha < m$, $m \in \mathbb{N}$, and let $0 < a < u < 1$. If ϕ is smooth on $[G(a), G(u)]$, then*

$${}^C D_{a+}^\alpha (\phi \circ G)(u) = \frac{1}{\Gamma(m-\alpha)} \int_{G(a)}^{G(u)} (u - F(x))^{m-\alpha-1} (\mathcal{L}_F^m \phi)(x) F'(x) dx. \quad (55)$$

The endpoint $a = 0$ is valid whenever the displayed integral converges.

Proof. Insert (51) into the Caputo definition (15) and substitute $v = F(x)$. $\square$

For integer m , (51) itself gives the complete formula for the m th derivative of any composite $\phi(G(u))$; for an n -fold antiderivative, [Theorem 4.1](#) with $\beta = n$ gives the corresponding complete formula. Thus the two transmutations cover all integer and noninteger orders on both sides of order zero.

5.3 Abel inversion identity

If $\phi(0) = 0$ and the relevant operators exist, the standard identity $I_{0+}^\alpha {}^C D_{0+}^\alpha (\phi \circ G) = \phi \circ G$ becomes

$$I_{0+}^\alpha \left[\frac{1}{\Gamma(1-\alpha)} \int_0^{G(v)} \phi'(x)(v - F(x))^{-\alpha} dx \right] (u) = \phi(G(u)). \quad (56)$$

This is an Abel-type inversion formula whose data are expressed entirely through the forward function F . It suggests stable numerical schemes for G that avoid differentiating a numerically inverted CDF.

5.4 Endpoint asymptotics

Set

$$q = G(u), \quad \rho(u) := \frac{qF'(q)}{F(q)} = \frac{G(u)}{uG'(u)}. \quad (57)$$

The sharp endpoint analysis in the audited repository [11], combined with standard regular-variation language [3], implies

$$\log \frac{1}{G(u)} \sim \sqrt{2 \log 2 \log \frac{1}{u}}, \quad \rho(u) \sim \sqrt{\frac{2 \log(1/u)}{\log 2}}. \quad (58)$$

In particular, G is slowly varying at zero: for every fixed $t > 0$, $G(tu)/G(u) \rightarrow 1$ as $u \downarrow 0$.

Proposition 5.4 (Leading fractional primitives). *For fixed $\beta > 0$ and $p > 0$,*

$$I_{0+}^\beta G^p(u) \sim \frac{u^\beta G(u)^p}{\Gamma(\beta + 1)}, \quad u \downarrow 0. \quad (59)$$

Proof. After $v = ut$,

$$I_{0+}^\beta G^p(u) = \frac{u^\beta G(u)^p}{\Gamma(\beta)} \int_0^1 (1-t)^{\beta-1} \frac{G(ut)^p}{G(u)^p} dt. \quad (60)$$

The uniform convergence theorem and Potter bounds for slowly varying functions give dominated convergence. The remaining integral is $1/\beta$. $\square$

Proposition 5.5 (Caputo endpoint laws). *For fixed $0 < \alpha < 1$ and $p > 0$,*

$${}^C D_{0+}^\alpha G^p(u) \sim \frac{u^{-\alpha} G(u)^p}{\Gamma(1-\alpha)}, \quad u \downarrow 0. \quad (61)$$

For $p = 1$, writing $s = 1 - u$,

$${}^C D_{0+}^\alpha G(1-s) \sim \Gamma(\alpha) \frac{s^{-\alpha} G(s)}{\rho(s)}, \quad s \downarrow 0. \quad (62)$$

Proof architecture. The first assertion is the order-zero boundary case of the monotone-density theorem for fractional derivatives of slowly varying functions. It can also be derived from the cancellation-safe Marchaud form

$${}^C D_{0+}^\alpha f(u) = \frac{u^{-\alpha}}{\Gamma(1-\alpha)} \left[f(u) + \alpha \int_0^1 \frac{f(u) - f(ut)}{(1-t)^{\alpha+1}} dt \right]. \quad (63)$$

For the right endpoint, symmetry gives

$${}^C D_{0+}^\alpha G(1-s) = \frac{1}{\Gamma(1-\alpha)} \int_s^1 (r-s)^{-\alpha} G'(r) dr. \quad (64)$$

Now $G'(r) = r^{-1}G(r)/\rho(r)$ is regularly varying with index -1 . Scaling $r = sy$ gives $\int_1^\infty (y-1)^{-\alpha} y^{-1} dy = \Gamma(1-\alpha)\Gamma(\alpha)$. $\square$

Theorem 5.6 (Fractional integrability frontier). *For $0 < \alpha < 1$ and $0 < r < \infty$,*

$${}^C D_{0+}^\alpha G \in L^r(0, 1) \iff r \leq \frac{1}{\alpha}. \quad (65)$$

The same threshold holds for G^p with any fixed $p > 0$.

Proof. At either endpoint, Theorem 5.5 is a power $u^{-\alpha}$ times a slowly varying factor. For $\alpha r < 1$ the power is integrable; for $\alpha r > 1$ no slowly varying factor can repair the power divergence. At the critical exponent $r = 1/\alpha$, use (58): after $L = \log(1/u)$, the integral is bounded by an integral of $\exp(-c\sqrt{L})$ times a power of L , which converges. $\square$

The critical endpoint is unusually included. This is a direct consequence of the stretched-exponential-in- $\sqrt{\log}$ decay of G , and would fail for many more slowly decaying quantiles.

5.5 First correction and the log-periodic phase

The leading term uses only slow variation. A first correction requires a quantitative de Haan expansion. The repository's sharp endpoint theory contains a nonconstant dyadic periodic phase, so it is important not to hide that phase inside an unjustified constant.

Conditional theorem 5.7 (First fractional correction). *Assume that, in each of the weighted L^1 spaces used below,*

$$\frac{G(ut)}{G(u)} = 1 + \frac{\log t}{\rho(u)} + \mathcal{O}\left(\frac{(1 + |\log t|)^2}{\rho(u)^2}\right) \quad (66)$$

with an error uniform as $u \downarrow 0$. Then

$$\frac{\Gamma(\beta + 1)I_{0+}^\beta G^p(u)}{u^\beta G(u)^p} = 1 - \frac{p[\psi(\beta + 1) + \gamma]}{\rho(u)} + \mathcal{O}(\rho(u)^{-2}), \quad (67)$$

where ψ is the digamma function and γ is Euler's constant. Moreover,

$$\frac{\Gamma(1 - \alpha)u^\alpha {}^C D_{0+}^\alpha G^p(u)}{G(u)^p} = 1 + \frac{p[-\psi(1 - \alpha) - \gamma]}{\rho(u)} + \mathcal{O}(\rho(u)^{-2}). \quad (68)$$

Proof architecture. For (67), insert (66) after $v = ut$. The coefficient is the beta expectation

$$\beta \int_0^1 (1 - t)^{\beta-1} \log t \, dt = -\psi(\beta + 1) - \gamma. \quad (69)$$

For Caputo, use the Marchaud form. The coefficient is

$$\alpha \int_0^1 \frac{-\log t}{(1 - t)^{\alpha+1}} \, dt = -\psi(1 - \alpha) - \gamma. \quad (70)$$

$\square$

Conjecture 5.8 (Phase-refined de Haan expansion). *The Fabius inverse satisfies (66) with a complete asymptotic expansion in $\rho(u)^{-1}$ whose coefficients are polynomials in $\log t$ and smooth periodic functions of the exact lower-Lambert dyadic phase. The periodic dependence first enters beyond the universal coefficients displayed in (67) and (68).*

This conjecture is deliberately compatible with the repository's proof that the endpoint phase is nonconstant; it does not replace the phase by a scalar.

6 A complex-order generalized Stieltjes system

6.1 Definition and elementary structure

The generalized Stieltjes class and its positivity theory provide the natural ambient transform language [13]. Work first on the principal slit plane

$$\Omega := \mathbb{C} \setminus (-\infty, 1]. \quad (71)$$

For $z \in \Omega$, use the principal branch of $\text{Log}(z - x)$ simultaneously for all $x \in [0, 1]$ and define

$$\mathcal{S}_\lambda(z) := \int_0^1 (z - x)^{-\lambda} dF(x) = \int_0^1 (z - G(u))^{-\lambda} du. \quad (72)$$

The upper and lower banks of the slit supply the boundary values used below; other branches live naturally on the universal cover of $\mathbb{C} \setminus [0, 1]$. For each fixed $z \in \Omega$, $\lambda \mapsto \mathcal{S}_\lambda(z)$ is entire, because the compact family $\{\text{Log}(z - x) : 0 \leq x \leq 1\}$ is bounded. Parameter derivatives generate logarithmic kernels:

$$(-\partial_\lambda)^m \mathcal{S}_\lambda(z) = \int_0^1 (z - x)^{-\lambda} \text{Log}^m(z - x) dF(x). \quad (73)$$

For $z \in \Omega$ with $|z| > 1$ the binomial expansion gives

$$\mathcal{S}_\lambda(z) = z^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda)_n}{n!} \frac{d_n}{z^n}. \quad (74)$$

Every coefficient is rational and computable from (11). At negative integer orders, $\mathcal{S}_{-m}(z) = \mathbb{E}(z - X)^m$ is a polynomial; positive integer orders are the successive derivatives of the ordinary Stieltjes transform.

6.2 Dyadic order recurrence

Theorem 6.1 (Dyadic generalized-Stieltjes equations). *For all $z \in \Omega$ and all $\lambda \in \mathbb{C}$, by analytic continuation across the removable value $\lambda = 1$,*

$$\mathcal{S}_\lambda(z) = \frac{2^\lambda}{1 - \lambda} [\mathcal{S}_{\lambda-1}(2z) - \mathcal{S}_{\lambda-1}(2z - 1)]. \quad (75)$$

It also satisfies the same-order differential–dyadic equation

$$\partial_z \mathcal{S}_\lambda(z) = 2^{\lambda+1} [\mathcal{S}_\lambda(2z) - \mathcal{S}_\lambda(2z - 1)], \quad (76)$$

and the order-raising derivative identity

$$\partial_z \mathcal{S}_\lambda(z) = -\lambda \mathcal{S}_{\lambda+1}(z). \quad (77)$$

Proof. Using $X \stackrel{d}{=} (U + X')/2$,

$$\mathcal{S}_\lambda(z) = 2^\lambda \mathbb{E} \int_0^1 (2z - X' - u)^{-\lambda} du \quad (78)$$

$$= \frac{2^\lambda}{1 - \lambda} \mathbb{E} \left[(2z - X')^{1-\lambda} - (2z - 1 - X')^{1-\lambda} \right]. \quad (79)$$

This is (75). Differentiating it and using (77) gives (76); alternatively, differentiate under the defining integral and use the density refinement equation. $\square$

At $\lambda = 1$, the removable limit is the logarithmic identity

$$\mathcal{S}_1(z) = 2\mathbb{E} [\text{Log}(2z - X) - \text{Log}(2z - 1 - X)]. \quad (80)$$

For the centered variable Y , define

$$\mathcal{C}_\lambda(w) := \mathbb{E}(w - Y)^{-\lambda}. \quad (81)$$

Then

$$\mathcal{S}_\lambda(z) = 2^\lambda \mathcal{C}_\lambda(2z - 1), \quad \mathcal{C}'_\lambda(w) = 2^\lambda [\mathcal{C}_\lambda(2w + 1) - \mathcal{C}_\lambda(2w - 1)]. \quad (82)$$

The latter is a transform-level analogue of Rvachev's differential equation.

6.3 Laplace representation and exact fractional order shifts

For $\Re z > 1$ and $\Re \lambda > 0$, the standard Laplace–Gamma representation [14] gives

$$\mathcal{S}_\lambda(z) = \frac{1}{\Gamma(\lambda)} \int_0^\infty t^{\lambda-1} e^{-zt} M_X(t) dt, \quad (83)$$

$$M_X(t) = \mathbb{E} e^{tX} = e^{t/2} \widehat{\text{up}}(it/2). \quad (84)$$

The sinc product becomes an infinite product of hyperbolic sinc factors on the imaginary axis.

Let $A = -\partial_z$, and define A^ν on this Laplace class by multiplying the integrand by t^ν . This is the natural Weyl fractional power of the positive generator $-\partial_z$ on the right half-plane.

Theorem 6.2 (Fractional spectral order shift). *Whenever $\Re z > 1$, $\Re \lambda > 0$, and $\Re(\lambda + \nu) > 0$,*

$$A^\nu \mathcal{S}_\lambda(z) = \frac{\Gamma(\lambda + \nu)}{\Gamma(\lambda)} \mathcal{S}_{\lambda+\nu}(z). \quad (85)$$

In particular, for $n \in \mathbb{N}$,

$$\frac{d^n}{dz^n} \mathcal{S}_\lambda(z) = (-1)^n (\lambda)_n \mathcal{S}_{\lambda+n}(z), \quad (86)$$

and the n -fold ordinary primitive is

$$\int \cdots \int \mathcal{S}_\lambda(z) dz^n = (-1)^n \frac{\Gamma(\lambda - n)}{\Gamma(\lambda)} \mathcal{S}_{\lambda-n}(z) + P_{n-1}(z), \quad (87)$$

away from the exceptional integer parameters, with P_{n-1} an arbitrary polynomial of degree at most $n - 1$.

Proof. Apply the multiplier t^ν to (83) and compare with the same representation at order $\lambda + \nu$. $\square$

For $\nu > 0$, the inverse power $A^{-\nu}$ is a fractional antiderivative with boundary condition at $+\infty$:

$$(A^{-\nu} f)(z) = \frac{1}{\Gamma(\nu)} \int_z^\infty (w - z)^{\nu-1} f(w) dw. \quad (88)$$

Thus (85) gives an exact continuum of spectral derivatives and antiderivatives, not merely an interpolation of integer formulas.

6.4 Boundary jump as a fractional integral of the density

Theorem 6.3 (Fractional Plemelj–Abel jump). *Let $0 < \lambda < 1$ and $0 < x < 1$. The nontangential boundary values satisfy*

$$\mathcal{S}_\lambda(x + i0) - \mathcal{S}_\lambda(x - i0) = -2i \sin(\pi\lambda) \int_x^1 (t - x)^{-\lambda} F'(t) dt. \quad (89)$$

Equivalently,

$$I_{1-}^{1-\lambda} F'(x) = -\frac{\Gamma(\lambda)}{2\pi i} [\mathcal{S}_\lambda(x + i0) - \mathcal{S}_\lambda(x - i0)]. \quad (90)$$

Proof. For $t < x$, $x - t$ lies on the positive real axis and has no jump. For $t > x$,

$$(x - t + i0)^{-\lambda} - (x - t - i0)^{-\lambda} = -2i \sin(\pi\lambda) (t - x)^{-\lambda}. \quad (91)$$

Integrate against $F'(t) dt$ and use $\Gamma(\lambda)\Gamma(1 - \lambda) = \pi/\sin(\pi\lambda)$. $\square$

At $\lambda = 1$, the usual Stieltjes inversion recovers the density itself. For $0 < \lambda < 1$, the jump recovers a right-sided fractional integral of the density; applying a right Riemann–Liouville derivative then reconstructs F' . This gives a continuum of inversion formulas connecting complex boundary data to fractional real-variable smoothing.

6.5 Endpoint values and inverse moments

Because the Fabius density is flat at both endpoints, all inverse moments exist. The endpoint value

$$\mathcal{S}_\lambda(1) = \mathbb{E}(1 - X)^{-\lambda} = \mathbb{E}X^{-\lambda} = \int_0^1 G(u)^{-\lambda} du \quad (92)$$

therefore extends entire in λ . Formula (75), interpreted at endpoint limits, offers a dyadic recursion for these inverse moments. Combining it with (74) is a promising route to new q -series and q -binomial representations at dyadic spectral points.

7 Weyl and Riesz fractional calculus of Rvachev's function

7.1 Exact order-lowering dilation law

The ordinary Rvachev equation [1, 6] is

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1). \quad (93)$$

A naive attempt to replace the derivative by a fractional derivative while keeping up on the right is false. The correct equation lowers the order on the right-hand side.

Theorem 7.1 (Weyl order-lowering equation). *For every $\alpha \in \mathbb{C}$, as tempered distributions and with a consistent branch of $(i\xi)^\alpha$,*

$$D_+^\alpha \text{up}(x) = 2^\alpha [(D_+^{\alpha-1} \text{up})(2x + 1) - (D_+^{\alpha-1} \text{up})(2x - 1)]. \quad (94)$$

At $\alpha = 1$ this is (93); at integer n it gives all higher derivative equations.

Proof. The Fourier transform of the right-hand side is

$$2^\alpha i \sin(\xi/2) (i\xi/2)^{\alpha-1} \widehat{\text{up}}(\xi/2) = 2i \sin(\xi/2) (i\xi)^{\alpha-1} \widehat{\text{up}}(\xi/2) \quad (95)$$

$$= (i\xi)^\alpha \widehat{\text{up}}(\xi), \quad (96)$$

where the last equality uses $\widehat{\text{up}}(\xi) = \text{sinc}(\xi/2) \widehat{\text{up}}(\xi/2)$. $\square$

The theorem is the structurally correct fractional generalization: the self-similarity survives, but the dilation intertwines adjacent derivative orders. No contradiction with the repository's log-periodic obstruction arises, because the right side is not a scalar multiple of the original up.

7.2 One-sided Weyl tails

For $0 < \alpha < 1$, the left Marchaud–Weyl derivative is

$$D_+^\alpha f(x) = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty \frac{f(x) - f(x-t)}{t^{1+\alpha}} dt. \quad (97)$$

If $x > 1$, then $\text{up}(x) = 0$ and only the translated support contributes.

Theorem 7.2 (Convergent Weyl tail series). *For $0 < \alpha < 1$ and $x > 1$,*

$$D_+^\alpha \text{up}(x) = \frac{x^{-1-\alpha}}{\Gamma(-\alpha)} \sum_{j=0}^{\infty} \frac{(\alpha+1)_{2j}}{(2j)!} \frac{c_j}{x^{2j}}. \quad (98)$$

The series converges absolutely for every $x > 1$. In particular,

$$D_+^\alpha \text{up}(x) = \frac{x^{-1-\alpha}}{\Gamma(-\alpha)} \left[1 + \frac{(\alpha+1)(\alpha+2)}{18x^2} + \frac{19(\alpha+1)_4}{16200x^4} + \dots \right]. \quad (99)$$

Proof. For $x > 1$, (97) becomes

$$D_+^\alpha \text{up}(x) = \frac{1}{\Gamma(-\alpha)} \int_{-1}^1 (x-y)^{-1-\alpha} \text{up}(y) dy. \quad (100)$$

Expand $(1-y/x)^{-1-\alpha}$. Odd moments vanish and the even moments are c_j . Absolute convergence follows from $|y/x| < 1$. $\square$

Since $\Gamma(-\alpha) < 0$, the left Weyl tail is negative to the right of the support. The right Weyl derivative has the reflected tail to the left. Ordinary integer derivatives retain compact support, but every genuinely noninteger Weyl derivative develops an algebraic tail. This is a sharp, quantified form of nonlocality.

7.3 Riesz fractional-Laplacian tails

For $0 < \alpha < 2$, the one-dimensional fractional Laplacian can be normalized as

$$(-\Delta)^{\alpha/2} f(x) = C_\alpha \text{pv} \int_{\mathbb{R}} \frac{f(x) - f(y)}{|x-y|^{1+\alpha}} dy, \quad C_\alpha = \frac{\Gamma(1+\alpha) \sin(\pi\alpha/2)}{\pi}. \quad (101)$$

Theorem 7.3 (Riesz tail series). *For $0 < \alpha < 2$ and $|x| > 1$,*

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_\alpha |x|^{-1-\alpha} \sum_{j=0}^{\infty} \frac{(\alpha+1)_{2j}}{(2j)!} \frac{c_j}{|x|^{2j}}. \quad (102)$$

Hence

$$(-\Delta)^{\alpha/2} \text{up}(x) \sim -C_\alpha |x|^{-1-\alpha}. \quad (103)$$

Proof. Outside the support, $\text{up}(x) = 0$, so the principal value is unnecessary and (101) is minus a positive potential integral. Expand the kernel exactly as in Theorem 7.2. $\square$

The coefficients in both nonlocal tails are rational and obey the explicit recurrence (12). Thus a global fractional operation converts the compactly supported atomic function into an algebraically decaying function whose complete exterior expansion is arithmetically controlled by the same moment sequence that appears in the Fabius theory.

8 An entire-order hierarchy from Fabius differences

8.1 Autocorrelation and the difference density

Let X_1, X_2 be independent copies of X and put

$$D := X_2 - X_1 \in [-1, 1]. \quad (104)$$

Its density h is even. If

$$R(s) := \int_{\mathbb{R}} \text{up}(y) \text{up}(y + s) \, dy, \quad (105)$$

then scaling $D = (Y_2 - Y_1)/2$ gives

$$h(a) = 2R(2a). \quad (106)$$

Equivalently,

$$h(a) = \int_0^1 F'(x) F'(x + a) \, dx, \quad (107)$$

where F' is extended by zero outside $[0, 1]$. Thus h is C^∞ and flat at ± 1 .

8.2 Gamma-normalized positive-part order

For $a \in (-1, 1)$ and initially $\Re \beta > -1$, define

$$\mathcal{K}_\beta(a) := \frac{\mathbb{E}(D - a)_+^\beta}{\Gamma(\beta + 1)} = \frac{1}{\Gamma(\beta + 1)} \int_a^1 (d - a)^\beta h(d) \, dd. \quad (108)$$

Normalization warning 8.1. The Gamma factor is part of the definition. For example, $\mathbb{E}(D_+)^2 = 1/36$, but $\mathcal{K}_2(0) = \mathbb{E}(D_+)^2/\Gamma(3) = 1/72$. This normalization is what makes the order ladder coefficient-free.

Theorem 8.2 (Entire stop-loss order). *For every fixed $a \in (-1, 1)$, $\beta \mapsto \mathcal{K}_\beta(a)$ extends to an entire function. For every $N \geq 0$ and $\Re \beta > -N - 1$,*

$$\mathcal{K}_\beta(a) = \frac{(-1)^N}{\Gamma(\beta + N + 1)} \int_0^{1-a} t^{\beta+N} h^{(N)}(a + t) \, dt. \quad (109)$$

It satisfies the entire-order differential ladder

$$-\partial_a \mathcal{K}_\beta(a) = \mathcal{K}_{\beta-1}(a), \quad (110)$$

and at negative integers

$$\mathcal{K}_{-m}(a) = (-1)^{m-1} h^{(m-1)}(a), \quad m \geq 1. \quad (111)$$

Proof. Set $t = d - a$ in (108). Repeated integration by parts moves derivatives onto $h(a + t)$. Every upper boundary term vanishes because h is flat at 1; every lower term vanishes in the initial half-plane and is then encoded by the zeros of $1/\Gamma(\beta + 1)$. The resulting formulas (109) agree on overlaps and give an entire continuation. Differentiating in a proves (110); taking $N = m$ in (109) gives (111). $\square$

The order axis now has a concrete interpretation:

β	$\mathcal{K}_\beta(a)$
$\beta > 0$	fractional positive-part or stop-loss integral
0	$\mathbb{P}(D > a)$
-1	$h(a)$
-2	$-h'(a)$
-m	$(-1)^{m-1} h^{(m-1)}(a).$

(112)

Thus a single entire function of the order joins tail probabilities, autocorrelations, derivatives, and fractional antiderivatives.

8.3 Triangular dyadic refinement

From (7),

$$D \stackrel{d}{=} \frac{V + D'}{2}, \quad (113)$$

where $V = U_2 - U_1$ has triangular density $\tau(v) = 1 - |v|$ on $[-1, 1]$ and D' is an independent copy.

Theorem 8.3 (Entire-order triangular refinement). *For every $a \in \mathbb{R}$ and initially $\Re\beta > -1$,*

$$\mathcal{K}_\beta(a) = 2^{-\beta} \int_{-1}^1 (1 - |v|) \mathcal{K}_\beta(2a - v) dv. \quad (114)$$

For fixed interior a , the identity extends to every $\beta \in \mathbb{C}$ by Theorem 8.2.

Proof. Condition on $V = v$ in (113):

$$(D - a)_+^\beta = 2^{-\beta} (D' - (2a - v))_+^\beta. \quad (115)$$

Integrate against the triangular density and divide by $\Gamma(\beta + 1)$. $\square$

At $\beta = -1$, (114) is the density refinement equation; at lower negative integers it gives all derivative refinements. At positive orders it supplies a stable recursive equation for fractional stop-loss integrals.

8.4 A double inverse-Fabius integral

Using $X_j = G(U_j)$ with independent uniforms,

$$\mathcal{K}_\beta(a) = \frac{1}{\Gamma(\beta + 1)} \int_0^1 \int_0^1 (G(v) - G(u) - a)_+^\beta du dv. \quad (116)$$

This is a definite integral involving two inverse Fabius functions, while (106) expresses the same hierarchy through Rvachev autocorrelation and (107) through the forward density. It is therefore a literal three-function bridge.

At $a = 0$ and $\beta = 1$, the Gini mean-difference identity gives the mixed forward-function integral

$$\mathcal{K}_1(0) = \frac{1}{2} \mathbb{E}|X_2 - X_1| = \int_0^1 F(x)(1 - F(x)) dx. \quad (117)$$

Numerically this constant is 0.0955632320160...

8.5 Fractional Fourier energy

The characteristic function of D is

$$\varphi_D(t) = |\varphi_X(t)|^2 = \widehat{\text{up}}(t/2)^2. \quad (118)$$

The standard fractional-moment identity [12, 10] now gives a transform formula.

Theorem 8.4 (Fractional sinc-product energy). *For $0 < \beta < 2$,*

$$\mathcal{K}_\beta(0) = \frac{\sin(\pi\beta/2)}{\pi} \int_0^\infty \frac{1 - \widehat{\text{up}}(t/2)^2}{t^{1+\beta}} dt. \quad (119)$$

In particular,

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{\pi} \int_0^\infty \frac{1 - \widehat{\text{up}}(t/2)^2}{t^2} dt. \quad (120)$$

Proof. For $0 < \beta < 2$,

$$|x|^\beta = \frac{2\Gamma(\beta + 1) \sin(\pi\beta/2)}{\pi} \int_0^\infty \frac{1 - \cos(tx)}{t^{1+\beta}} dt. \quad (121)$$

Take expectation with $x = D$, use symmetry $\mathbb{E}(D_+)^{\beta} = \frac{1}{2} \mathbb{E}|D|^{\beta}$, and divide by $\Gamma(\beta + 1)$. $\square$

8.6 Exact even orders

For $n \geq 0$, independence and $D = (Y_2 - Y_1)/2$ give

$$\mathcal{K}_{2n}(0) = \frac{1}{2(2n)!4^n} \sum_{k=0}^n \binom{2n}{2k} c_k c_{n-k}. \quad (122)$$

The first values are

$$\mathcal{K}_2(0) = \frac{1}{72}, \quad (123)$$

$$\mathcal{K}_4(0) = \frac{11}{64800}, \quad (124)$$

$$\mathcal{K}_6(0) = \frac{211}{171460800}, \quad (125)$$

$$\mathcal{K}_8(0) = \frac{31657}{5246700480000}, \quad (126)$$

$$\mathcal{K}_{10}(0) = \frac{7647271}{354246723008640000}. \quad (127)$$

These rational constants are new mixed integral evaluations; by (116), each is a double integral of an even power of a difference of inverse Fabius values over the upper triangle $v > u$.

9 Numerical experiments

9.1 Reconstruction method

The accompanying script `numerical_experiments.py` reconstructs up directly from (8). On a periodic interval of length 4 with $N = 2^{17}$ Fourier modes it evaluates

$$\prod_{j=1}^{35} \text{sinc}(\xi/2^j) \quad (128)$$

and applies an inverse FFT. The interval is wider than the exact support, so there is no wrap-around overlap. The CDF is obtained by cumulative quadrature and the inverse G by monotonicity-preserving PCHIP interpolation. Independent exact recurrences generate d_n and c_n as rational numbers.

The principal sanity checks, including the exact dyadic value $F(1/4) = 5/72$ [7, 2, 11], are

$$\int_{-1}^1 \text{up} = 1, \quad \max \text{up} = 1, \quad \int_{-1}^1 y^2 \text{up}(y) dy = \frac{1}{9}, \quad F\left(\frac{1}{4}\right) = \frac{5}{72}. \quad (129)$$

At full resolution, the computed value of $F(1/4)$ differs from $5/72$ by about 1.6×10^{-10} .

9.2 Independent checks

For the quantile theorem, the left side is integrated directly in u and the right side directly in x ; for negative powers, the regularized entire continuation is compared with the direct inverse integral. Caputo values are checked against a cancellation-safe Marchaud formula that uses only values of G , not the transmuted x integral. The generalized Stieltjes equations are checked by density quadrature, and the moment series uses exact rational d_n . Difference refinements use high-accuracy weighted quadrature, while the Fourier-energy check uses the sinc product independently.

Table 1: Independent checks of the quantile–Volterra and Caputo transmutations.

case	direct form	transmuted form	relative error
$u = 0.2, \beta = 0.6, p = 1.0$	0.1226638614	0.1226638614	3.02×10^{-13}
$u = 0.5, \beta = 1.4, p = 2.3$	0.0274018083	0.0274018083	2.1×10^{-13}
$u = 0.8, \beta = 0.75, p = 0.4$	0.674404041	0.674404041	1.81×10^{-15}
$u = 0.5, \beta = 0.8, p = -0.7$	1.280994219	1.280994219	2.86×10^{-13}
$u = 0.8, \beta = 1.3, p = -1.2$	2.184447863	2.184447863	1.54×10^{-13}
$u = 0.2, \beta = 0.6, p = -0.4$	0.7110303503	0.7110303503	1.31×10^{-13}
$u = 0.6, \beta = 0.7, m = 1$	-0.7015705216	-0.7015705216	2.77×10^{-14}
$u = 0.6, \beta = 0.7, m = 2$	0.7090525688	0.7090525688	4.17×10^{-13}
$u = 0.6, \beta = 0.7, m = 3$	-0.8030874837	-0.8030874837	1.95×10^{-13}
$u = 0.2, \alpha = 0.3, p = 1.0$	0.504741341	0.504741341	6.35×10^{-12}
$u = 0.5, \alpha = 0.7, p = 1.0$	0.5952284038	0.5952284038	1.05×10^{-12}
$u = 0.8, \alpha = 0.45, p = 1.0$	0.6895063772	0.6895063773	1.46×10^{-10}

Table 2: Generalized Stieltjes equations and moment-series convergence.

z	λ	test	truncation	absolute residual
$1.7 + 0.4i$	$0.6 + 0.3i$	differential-dyadic	–	3.66×10^{-16}
$1.7 + 0.4i$	$0.6 + 0.3i$	order recurrence	–	5×10^{-16}
2.2	1.4	differential-dyadic	–	1.67×10^{-16}
2.2	1.4	order recurrence	–	3.33×10^{-16}
$-0.7 + 0.5i$	0.8	differential-dyadic	–	1.39×10^{-16}
$-0.7 + 0.5i$	0.8	order recurrence	–	1.39×10^{-15}
2.2	1.4	moment series	5	0.00154
2.2	1.4	moment series	10	7.87×10^{-6}
2.2	1.4	moment series	20	4.3×10^{-10}
2.2	1.4	moment series	30	3.87×10^{-14}
$1.7 + 0.4i$	$0.6 + 0.3i$	moment series	5	0.00207
$1.7 + 0.4i$	$0.6 + 0.3i$	moment series	10	2.08×10^{-5}
$1.7 + 0.4i$	$0.6 + 0.3i$	moment series	20	6.92×10^{-9}
$1.7 + 0.4i$	$0.6 + 0.3i$	moment series	30	4.62×10^{-12}

Table 3: Nonlocal tails and the entire-order difference hierarchy.

test	parameters	direct	predicted	relative error
Weyl tail	$x = 1.2, \alpha = 0.4$	-0.2434690215	-0.2434690025	7.8×10^{-8}
Weyl tail	$x = 1.5, \alpha = 0.8$	-0.09670592448	-0.09670592447	7.42×10^{-11}
Weyl tail	$x = 2.0, \alpha = 0.6$	-0.09489107745	-0.09489107745	2.63×10^{-11}
K_β refinement	$a = -0.2, \beta = 0.4$	0.5086799254	0.5086799254	1.79×10^{-14}
K_β refinement	$a = 0.0, \beta = 0.8$	0.1364947268	0.1364947268	9.35×10^{-14}
K_β refinement	$a = 0.25, \beta = 1.3$	0.00866136247	0.00866136247	4.45×10^{-14}
K_β refinement	$a = 0.5, \beta = 0.6$	0.002613518484	0.002613518484	8.63×10^{-15}
fractional energy	$\beta = 0.4$	0.2686797068	0.2686797079	3.99×10^{-9}
fractional energy	$\beta = 0.8$	0.1364947268	0.1364947274	4.11×10^{-9}
fractional energy	$\beta = 1.0$	0.09556323202	0.09556323241	4.11×10^{-9}
fractional energy	$\beta = 1.5$	0.03746967245	0.0374696726	4.11×10^{-9}

Table 4: Exact Gamma-normalized even values of the difference hierarchy.

order	exact $K_{2n}(0)$	numerical value
2	$\frac{1}{72}$	0.0138888888889
4	$\frac{64800}{211}$	0.00016975308642
6	$\frac{171460800}{31657}$	$1.23060198016 \times 10^{-6}$
8	$\frac{5246700480000}{7647271}$	$6.03369681967 \times 10^{-9}$
10	$\frac{354246723008640000}{}$	$2.15874149378 \times 10^{-11}$

Table 5: Approach to the leading slowly-varying fractional-integral asymptotic for $\beta = 0.7, p = 1$.

u	$\Gamma(\beta + 1)I^\beta G(u)/(u^\beta G(u))$
0.01	0.8642340122
0.0001	0.8928590708
10^{-6}	0.9076622860

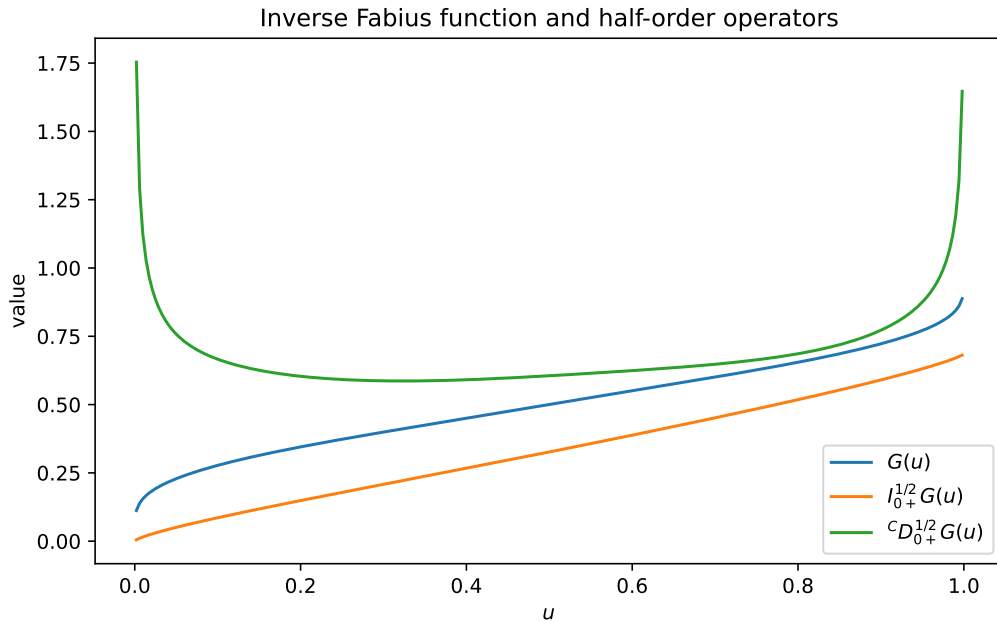


Figure 1: The inverse Fabius function together with its half-order Riemann–Liouville integral and half-order Caputo derivative. The Caputo operator amplifies endpoint structure, whereas the fractional integral smooths it.

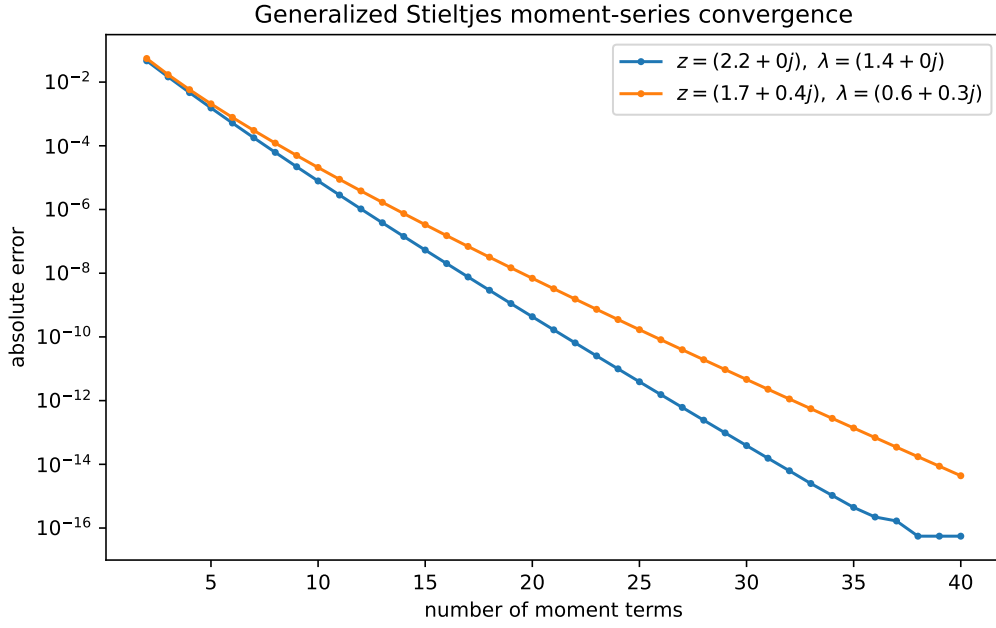


Figure 2: Convergence of the exact-moment expansion (74) at two real/complex parameter pairs.

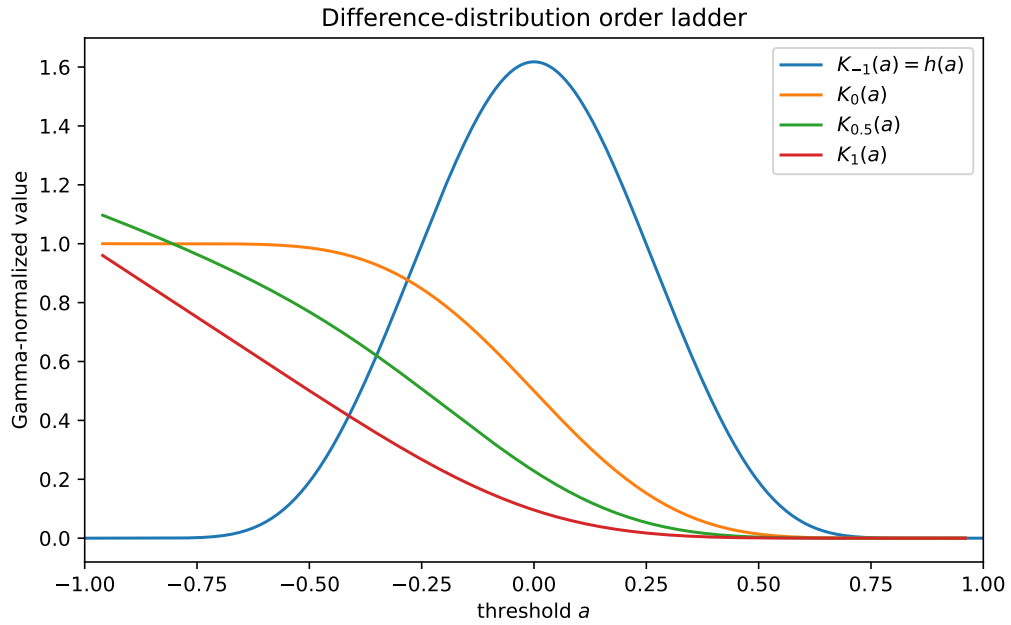


Figure 3: Several real-order slices of the entire difference hierarchy. The order -1 slice is the difference density, order 0 is its tail, and positive orders are fractional positive-part integrals.

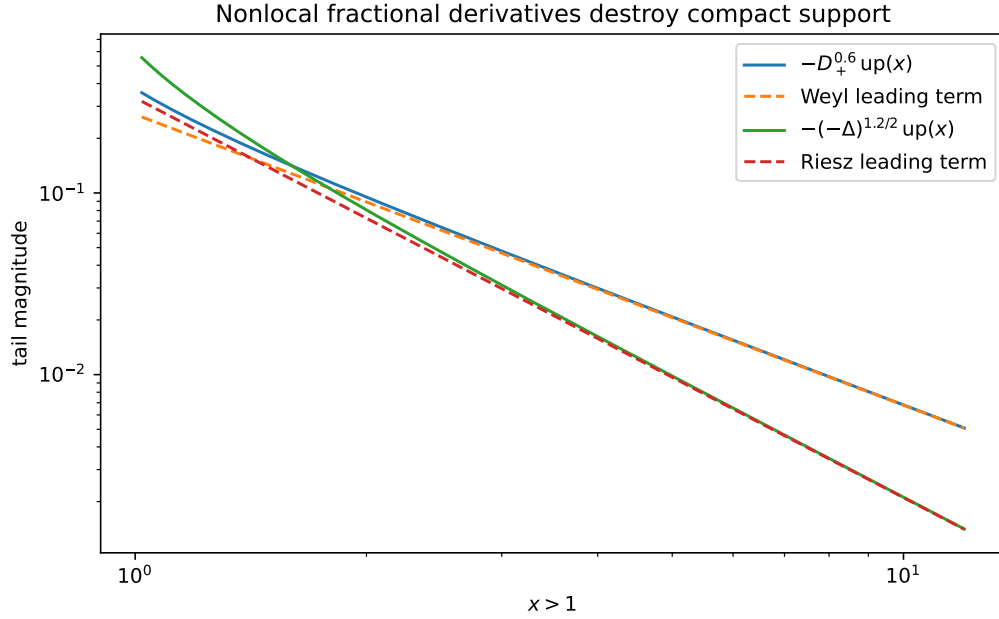


Figure 4: Weyl and Riesz fractional derivatives outside the support of u_p . Dashed curves are their leading $x^{-1-\alpha}$ terms. The agreement illustrates how noninteger differentiation creates exact algebraic tails.

9.3 Boundary jump experiment

At $x = 0.4$ and $\lambda = 0.35$, the predicted jump from (89) was compared with $\mathcal{S}_\lambda(x + i\varepsilon) - \mathcal{S}_\lambda(x - i\varepsilon)$. The absolute discrepancies for $\varepsilon = 10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}$ were, respectively,

$$5.89 \cdot 10^{-2}, \quad 1.80 \cdot 10^{-2}, \quad 6.01 \cdot 10^{-3}, \quad 1.81 \cdot 10^{-3}, \quad 6.02 \cdot 10^{-4}. \quad (130)$$

The regular decrease is consistent with convergence to the fractional Plemelj boundary value. This is a deliberately harsher test than the off-cut functional equations, because it resolves a singular limit with a finite Fourier grid.

10 Further deductions and research directions

10.1 All-order endpoint expansions

Theorem 5.7 shows that once a weighted de Haan expansion for $G(ut)/G(u)$ is known, fractional integration converts powers of $\log t$ to digamma and polygamma constants. At higher orders one expects complete Bell polynomials in derivatives of the beta function. The missing ingredient is not fractional calculus but uniform control of the exact dyadic endpoint phase.

Problem 10.1 (Phase-aware Watson lemma). Prove a Watson–de Haan lemma for slowly varying functions whose second-order auxiliary function contains a smooth periodic term of a lower-Lambert phase, and apply it simultaneously to Riemann–Liouville and Caputo kernels.

A successful theorem would mechanically generate all correction terms in (59) and (61), with coefficients expressible through Bell polynomials and polygamma values. This is a natural meeting point of the repository’s Lambert- W , Bell-polynomial, and fractional calculus strands.

10.2 A two-dimensional holonomic-like lattice

The pair of equations

$$\partial_z \mathcal{S}_\lambda = -\lambda \mathcal{S}_{\lambda+1}, \quad \mathcal{S}_\lambda(z) = \frac{2^\lambda}{1-\lambda} [\mathcal{S}_{\lambda-1}(2z) - \mathcal{S}_{\lambda-1}(2z-1)] \quad (131)$$

forms a lattice in continuous order λ and dyadically transformed spectral position z . The fractional shift theorem (85) fills the gaps between integer lattice levels.

Problem 10.2 (Special spectral points). Determine whether $\mathcal{S}_\lambda(z)$ at dyadic z admits convergent q -binomial, continued-fraction, or Mahler-type representations that extend the known rational moment and dyadic-value formulas. Particularly natural points are $z = 2^m$, $z = 1 + 2^{-m}$, and the centered points $w = 2z - 1$ fixed or periodic under $w \mapsto 2w \pm 1$.

Conjecture 10.3 (Dyadic resolvent rigidity). *For each real $\lambda > 0$, the differential–dyadic equation (76), the asymptotic $\mathcal{S}_\lambda(z) \sim z^{-\lambda}$ at infinity, and the generalized Stieltjes positivity conditions uniquely characterize $\mathcal{S}_\lambda$.*

This would be a transform-side uniqueness theorem parallel to the uniqueness of the original Rvachev function.

10.3 Zeros in the order plane

For fixed $a \in (-1, 1)$, $\mathcal{K}_\beta(a)$ is entire in β , and its values at negative integers are signed derivatives of h . The zeros in the order plane therefore interpolate the oscillatory geometry of every derivative of the autocorrelation density.

Conjecture 10.4 (Symmetry-forced order zeros). *Evenness gives the exact lattice of forced zeros $\mathcal{K}_{-2m}(0) = 0$ for $m \geq 1$. These zeros are eventually simple, and any additional sufficiently far-left real zeros interlace the forced lattice. A more flexible version asks whether the Jensen polynomials of a suitably shifted and rescaled order function become hyperbolic asymptotically.*

This is speculative, but it is numerically accessible because (109) remains stable far to the left after choosing N large enough.

10.4 Spectral analysis of the triangular refinement operator

For each order define

$$(\mathcal{T}_\beta f)(a) := 2^{-\beta} \int_{-1}^1 (1 - |v|) f(2a - v) dv. \quad (132)$$

Then $\mathcal{K}_\beta$ is a fixed point of $\mathcal{T}_\beta$. The order enters both the scalar prefactor and the regularity class selected by the boundary conditions.

Problem 10.5 (Order-dependent refinement spectrum). Develop the spectrum of $\mathcal{T}_\beta$ on weighted Sobolev, Mellin, or entire-function spaces. Determine whether analytic perturbation in β controls the zero set and growth of $\mathcal{K}_\beta(a)$.

The triangular kernel has Fourier transform $\text{sinc}^2(\xi/2)$, so this problem may connect the squared sinc product for the difference distribution to the original single sinc product for up.

10.5 Higher-dimensional differences and B-spline noise

For independent Fabius variables and coefficients $\varepsilon_j \in \{\pm 1\}$, consider

$$D_m := \sum_{j=1}^m \varepsilon_j X_j. \quad (133)$$

The fixed-point noise becomes a compact B-spline distribution obtained from a signed sum of uniforms. The same construction gives an entire hierarchy

$$\mathcal{K}_{m,\beta}(a) := \frac{\mathbb{E}(D_m - a)_+^\beta}{\Gamma(\beta + 1)} \quad (134)$$

and a refinement equation whose kernel is that B-spline. Even orders are finite polynomials in the rational moments d_n or c_n . This is a broad family of exact definite integrals involving several inverse Fabius functions.

10.6 Fractional operators and q-binomial arithmetic

The exact moment series in (74), the exterior series (98), and the exact even values (122) are all built from rational moment sequences. Those sequences are already connected in the repository to q -Pochhammer symbols and Gaussian binomial coefficients at $q = 2$ or $q = 1/2$.

Problem 10.6 (Arithmetic compression). Find direct q -hypergeometric forms for

$$\sum_{n \geq 0} \frac{(\lambda)_n}{n!} d_n z^{-n}, \quad \sum_{j \geq 0} \frac{(\alpha + 1)_{2j}}{(2j)!} c_j x^{-2j}, \quad (135)$$

possibly after extracting the dominant lower-Lambert phase or applying a Borel transform. Determine denominator bounds for their Taylor coefficients as functions of the order parameters.

10.7 Formalization roadmap

The most direct Lean formalization order is:

1. prove the layer-cake form of [Theorem 4.1](#) first for nonnegative ϕ' and real $\beta > 0$, then extend by signed measures;
2. formalize the operator identity $(D_u^m(\phi \circ G)) \circ F = \mathcal{L}_F^m \phi$ on compact subintervals of $(0, 1)$;
3. derive the Caputo transmutation from change of variables;
4. define $\mathcal{S}_\lambda$ initially for real $\lambda > 0$ and real $z > 1$, where all kernels are positive, then add complex analyticity;
5. formalize $\mathcal{K}_\beta$ for real $\beta > -1$, its refinement equation, and the integer negative-order identities before attempting entire continuation;
6. use the already formalized sinc product and moment recurrences for the Weyl/Riesz exterior expansions.

The first, second, and fifth steps require mostly measure theory, ordinary calculus, and Gamma identities. They should be substantially easier than the sharp endpoint asymptotic development already present in the repository.

11 Conclusion

The integral theory of the Fabius and Rvachev functions becomes much more coherent when the integration order itself is treated as a variable. Three operator families organize most of the new structure:

1. the quantile–Volterra kernel $(u - F(x))^\beta / \Gamma(\beta + 1)$, which turns arbitrary-order primitives and Caputo derivatives of G into forward-function integrals;
2. the complex-order resolvent $\mathcal{S}_\lambda(z)$, whose spectral derivatives, dyadic refinements, moment expansions, and boundary jumps are all order shifts of one another;
3. the entire positive-part hierarchy $\mathcal{K}_\beta(a)$, which joins inverse-Fabius double integrals, Fabius density correlations, Rvachev autocorrelations, fractional energies, and all derivatives of the difference density.

The Weyl/Riesz analysis adds a complementary lesson: compact support is an integer-order phenomenon. Noninteger differentiation immediately produces algebraic tails, but the tails are not mysterious; every coefficient is an exact rational Rvachev moment. Together, these formulas connect the repository’s existing themes – dyadic refinement, sinc products, rational moments, inverse asymptotics, Bell polynomials, Lambert W , and q -arithmetic – through a common continuous order parameter.

A Proof details for analytic continuation

A.1 Why the reciprocal Gamma factor is canonical

Let $g \in C^\infty[0, L]$ be flat at L , and define initially for $\Re \beta > -1$

$$M_g(\beta) := \frac{1}{\Gamma(\beta + 1)} \int_0^L t^\beta g(t) dt. \quad (136)$$

Repeated integration by parts gives

$$M_g(\beta) = \frac{(-1)^N}{\Gamma(\beta + N + 1)} \int_0^L t^{\beta+N} g^{(N)}(t) dt. \quad (137)$$

This is entire in β . At $\beta = -m$,

$$M_g(-m) = (-1)^{m-1} g^{(m-1)}(0). \quad (138)$$

Both [Theorem 8.2](#) and the ladders of [Section 3](#) are instances of this elementary lemma. The zeros of $1/\Gamma(\beta + 1)$ are not an ornamental normalization: they cancel the poles of the local Mellin transform generated by the Taylor series of g at zero.

A.2 Complex powers and branches

In the quantile formulas, $u - F(x) \geq 0$ and $x > 0$, so $(u - F(x))^\beta$ and x^p use real logarithms on the open integration interval. In the generalized Stieltjes transform, $z - x$ moves in a compact subset of the principal cut plane when $z \in \Omega$; a single branch of Log is fixed for all $x \in [0, 1]$. The upper and lower banks define the Plemelj values on $(0, 1)$. Statements on other cuts are branch-local and do not assert monodromy-free continuation around the support.

B Moment recurrences and exact constants

From $X = (U + X')/2$,

$$2^n d_n = \sum_{k=0}^n \binom{n}{k} \frac{d_k}{n - k + 1}. \quad (139)$$

Separating the $k = n$ term gives (11). The first values are

$$d_0 = 1, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{5}{18}, \quad d_3 = \frac{1}{6}, \quad d_4 = \frac{143}{1350}, \quad d_5 = \frac{19}{270}, \quad d_6 = \frac{1153}{23814}. \quad (140)$$

Similarly, $Y = (V + Y')/2$ and $\mathbb{E}V^{2m} = 1/(2m + 1)$ give

$$4^n c_n = \sum_{k=0}^n \binom{2n}{2k} \frac{c_k}{2(n - k) + 1}. \quad (141)$$

After separating $k = n$ and using

$$\frac{1}{2(n - k) + 1} \binom{2n}{2k} = \frac{1}{2n + 1} \binom{2n + 1}{2k}, \quad (142)$$

one obtains (12).

For the difference moments,

$$\mathbb{E}D^{2n} = \frac{1}{4^n} \sum_{k=0}^n \binom{2n}{2k} c_k c_{n-k}. \quad (143)$$

Symmetry contributes the factor $1/2$ for the positive half, and the Gamma normalization contributes $(2n)!$, giving (122).

C Reproducibility notes

The archive accompanying this report contains:

- `fabius_fractional_transform_frontiers.tex`, the source of this report;
- `fabius_fractional_transform_frontiers.pdf`, the rendered report;
- `numerical_experiments.py`, the fully commented experiment script;
- `numerical_results.txt` and `numerical_results.tex`, generated numerical output;
- four figures in both PDF and PNG forms;
- `README.md`, `CORPUS_AUDIT.md`, and `SHA256SUMS`.

The numerical script requires Python 3 with NumPy, SciPy, mpmath, and Matplotlib. Running

```
python numerical_experiments.py
latexmk -pdf fabius_fractional_transform_frontiers.tex
```

regenerates the tables, figures, and PDF. The `-quick` option uses a 2^{15} -point grid for a fast smoke test; the reported values use 2^{17} .

D Corpus novelty matrix

Topic	Prior repository coverage	Contribution of this report
Polynomial and monomial primitives	Extensive formulas for integer powers, negative powers, and dyadic reduction	Used only as baseline; not rederived in detail
Fractional primitives of F and up	Causal fractional integrals and Mellin/moment-tail formulas	Gamma-normalized entire-order packaging and connection to nonlocal exterior tails
Inverse Fabius integrals	Integer layer-cake formulas, beta transforms, inverse moments	Arbitrary complex-order quantile–Volterra theorem; entire power parameter; logarithmic and exponential families
Fractional derivatives of G	No Caputo transmutation found	Exact orders $(0, 1)$ and arbitrary orders through $\mathcal{L}_F^m$; endpoint and L^r theory
Stieltjes transforms	Ordinary Cauchy/Stieltjes transform and a dyadic equation	Complex order, order recurrence, fractional spectral shifts, moment series, and fractional boundary inversion
Fractional derivatives of up	Fourier norm spectrum and warning against naive scaling	Exact Weyl order-lowering dilation; convergent Weyl/Riesz algebraic tails
Autocorrelation and differences	Basic correlation identities and selected norms	Entire positive-part order, negative-order derivatives, triangular refinement, inverse double integrals, and fractional energy
Endpoint asymptotics	Sharp lower-Lambert and nonconstant periodic phase	Slow-variation consequences for fractional operators; phase-aware conditional correction and conjecture

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